Data Mining Course Project

Knowledge Discovery in Banking Datasets

Group of 5 Students, 14-Week Project

1. Project Overview

Students will apply the KDD (Knowledge Discovery in Databases) process end-to-end on real banking data to discover hidden patterns, segments, and association rules. Each group is assigned a different banking dataset covering domains such as credit risk, fraud, mortgage, customer churn, anti-money laundering, and institutional financial health. While datasets differ, all groups execute the same five-phase methodology.

The unifying theme across all groups is the banking domain. The unifying methodology is the KDD pipeline. The goal is discovery and interpretation of hidden knowledge, not prediction accuracy.

Course	Data Mining
Group Size	5 Students per Group
Duration	14 Weeks
Dataset	Each group is assigned a different banking dataset. See Dataset Reference Document.

2. Role Assignment

Each student holds a **primary** role throughout the project. However, each member should still contribute in each phase, especially in Phase 1 and Phase 5.

Role	**Responsibility**
Data Engineer (2 students)	Data collection, cleaning, integration, and preprocessing pipeline
Pattern Analyst (1 student)	Association rule mining and frequent itemset discovery
Segmentation Specialist (1 student)	Clustering analysis and entity profiling
Insight Communicator (1 student)	Visualization, storytelling, final report, and presentation

3. Project Phases

All groups execute the same 5 phases regardless of which dataset they are assigned. This ensures consistent assessment criteria and enables meaningful cross-group comparison at the end of the semester.

Phase 1: Data Understanding and Preprocessing

Owner	Data Engineer
Tasks	
	Explore the dataset: distributions, missing values, data types, and outliers
	Apply data cleaning: handle nulls, fix inconsistencies, remove duplicates
	Perform data transformation: normalization, encoding, and binning of continuous variables
	Conduct feature selection using correlation analysis and entropy measures
	Deliverable: Clean dataset file through rigorous data pipeline and Preprocessing Report
Data Mining Concepts	Data quality, ETL pipeline, attribute types, noise handling, data transformation
Phase Goal	Produce a clean, analysis-ready dataset and document every preprocessing decision with justification. This phase establishes the factual foundation for all subsequent mining steps.

Phase 2: Segmentation via Clustering

Owner	Segmentation Specialist
Tasks	
	Apply K-Means clustering to segment entities by behavioral or financial attributes
	Use DBSCAN to identify outlier profiles and noise points within the data
	Apply Hierarchical clustering and compare dendrograms across linkage methods
	Determine optimal K using the Elbow Method and Silhouette Score
	Profile each cluster: describe the entity type it represents and its distinguishing characteristics
	Deliverable: Cluster profiles with full business interpretation
Data Mining Concepts	Unsupervised pattern discovery, distance metrics, cluster validity indices, dendrogram analysis
Phase Goal	Identify naturally occurring groupings within the data and translate each group into a meaningful, named profile that describes real-world behavior or financial characteristics.

Phase 3: Association Rule Mining

Owner	Pattern Analyst
Tasks	
	Discretize continuous variables into meaningful categorical groups
	Apply the Apriori algorithm to discover frequent itemsets across the dataset
	Generate association rules and compute Support, Confidence, and Lift for each
	Filter rules to retain only meaningful, non-trivial, high-lift findings
	Discover and document at least 10 rules with clear business interpretation
	Deliverable: Rule table with ranked rules and business commentary
DM Concepts	Frequent pattern mining, rule interestingness measures, Apriori algorithm, Support, Confidence, Lift
Phase Goal	Surface co-occurrence patterns in the data that reveal non-obvious relationships between attributes, findings that cannot be seen through simple tabulation or aggregation.

Phase 4: Anomaly and Outlier Detection

Owner	Data Engineer and Pattern Analyst
Tasks	
	Apply statistical methods (IQR, Z-score) to flag anomalous behaviors or values
	Use Isolation Forest as a structural anomaly discovery tool
	Cross-reference detected anomalies with cluster outliers from Phase 2
	Investigate each anomaly: data error, rare legitimate case, or potential risk signal?
	Deliverable: Anomaly report with flagged records, explanation, and business interpretation
Data Mining Concepts	Outlier analysis, statistical anomaly detection, Isolation Forest, anomaly typology
Phase Goal	Identify records that deviate substantially from the rest of the data and determine whether each deviation represents a data quality issue, an unusual but valid case, or a signal worth escalating.

Phase 5: Visualization and Knowledge Presentation

Owner	Insight Communicator
Tasks	

	Build an interactive dashboard using Google Looker Studio or Python Dash (Tableau Public and Power BI are also options)
	Visualize cluster maps, rule networks, outlier plots, and relevant distributions
	Write a Knowledge Discovery Report translating all findings into plain business language
	Answer the central question: What did we discover that was not already obvious from the raw data?
	Deliverable: Dashboard, 10-minute group presentation, and final written report
Data Mining Concepts	Data storytelling, knowledge presentation, dashboard design, business communication
Phase Goal	Communicate discovered knowledge to a non-technical audience clearly and convincingly. The measure of success is whether findings are actionable and non-trivial, not whether a model achieved high accuracy.

4. Tech Stack

Tool	Purpose
Python (pandas, mlxtend, scikit-learn)	Core analysis and mining algorithms
R	Core analysis and mining algorithms
Jupyter Notebook	Documentation and reproducible code
Matplotlib, Seaborn, Plotly	Data visualization and charts
Mage / Prefect / Airflow	Pipeline
Plotly Dash / Bokeh /	Dashboard
GitHub	Collaboration and version control

5. Grading Breakdown

Component	Weight	Evaluated On
Preprocessing	20%	Completeness, justification of choices
Clustering Analysis	20%	Validity metrics, business interpretation
Association Rule Mining	20%	Rule quality, non-triviality, lift values
Anomaly Detection	20%	Depth of investigation, explanations
Final Presentation and Dashboard	20%	Clarity, insight quality, storytelling

5.1 Preprocessing (15%)

Criterion	Excellent (85-100)	Good (75-84)	Average (65-74)	Poor (<65)
Data Cleaning	All missing values, duplicates, and inconsistencies handled with clear justification for each decision	Most issues addressed with adequate justification, minor gaps present	Basic cleaning done but justification is incomplete or inconsistent	Cleaning is superficial or missing; little to no documentation
Data Transformation	Normalization, encoding, and binning applied correctly and appropriately for the dataset	Transformations mostly correct with minor methodological issues	Some transformations applied but selection is poorly justified	Transformations missing or incorrectly applied
Feature Selection	Feature selection conducted using both correlation and entropy measures with findings clearly explained	Feature selection performed using at least one method with reasonable explanation	Feature selection attempted but method choice or interpretation is weak	No feature selection performed or rationale entirely absent

5.2 Clustering Analysis (25%)

Criterion	Excellent (85-100)	Good (75-84)	Average (65-74)	Poor (<65)
Algorithm Application	K-Means, DBSCAN, and Hierarchical clustering all applied correctly with appropriate parameter choices	At least two algorithms applied correctly, minor parameter issues	One algorithm applied correctly, others attempted with errors	Algorithms applied incorrectly or only one attempted
Optimal K Selection	Elbow Method and Silhouette Score both	Both methods used but interpretation	Only one method used, or	No validity measure used

	used, results interpreted correctly and used to justify final K	or justification is partially weak	interpretation is unclear	or K chosen arbitrarily
Cluster Profiling	Each cluster described with a named profile, distinguishing attributes, and meaningful business interpretation	Clusters described but business interpretation is generic or incomplete	Clusters labeled but profiles lack depth or domain relevance	No profiling done or descriptions are technically incorrect

5.3 Association Rule Mining (25%)

Criterion	Excellent (85-100)	Good (75-84)	Average (65-74)	Poor (<65)
Discretization	Continuous variables binned into meaningful, domain-relevant categories with clear rationale	Discretization done correctly but bin boundaries are not well justified	Discretization attempted but categories are arbitrary or too coarse	No discretization done or approach fundamentally incorrect
Rule Generation	Apriori applied correctly, Support, Confidence, and Lift computed and used to filter rules effectively	Rules generated correctly, filtering applied but threshold choices not fully explained	Rules generated but filtering is weak or interestingness measures misused	Rules not generated correctly or interestingness measures absent
Rule Interpretation	At least 10 non-trivial rules documented with specific, accurate, and actionable business interpretation	At least 10 rules documented but some interpretations are generic or partially inaccurate	Fewer than 10 rules or interpretations are vague and domain-disconnected	Rules listed without interpretation or interpretation is incorrect

5.4 Anomaly Detection (15%)

Criterion	Excellent (85-100)	Good (75-84)	Average (65-74)	Poor (<65)
Detection Methods	IQR, Z-score, and Isolation Forest all applied correctly and results compared systematically	At least two methods applied correctly with basic comparison	One method applied correctly, others attempted with errors	Only one method applied or methods used incorrectly
Cross-referencing	Detected anomalies cross-referenced with Phase 2 cluster outliers with clear reasoning	Cross-referencing attempted but reasoning is partial	Anomalies listed without reference to clustering results	No cross-referencing performed
Business Interpretation	Each flagged anomaly classified as data error, rare case, or risk signal with specific supporting evidence	Most anomalies interpreted but evidence is not consistently specific	Anomalies flagged but interpretation is generic or speculative	No interpretation provided or interpretations are incorrect

5.5 Final Presentation and Dashboard (20%)

Criterion	Excellent (85-100)	Good (75-84)	Average (65-74)	Poor (<65)
Dashboard Quality	Dashboard is interactive (<100ms), displays all required visualizations clearly, and is accessible to a non-technical audience	Dashboard functional and mostly complete, minor clarity or interactivity issues	Dashboard present but incomplete or difficult to interpret	No dashboard or visualizations are missing or non-functional

Knowledge Report	Report translates all findings into plain business language, answers the central discovery question directly and specifically	Report covers most findings adequately but central question is only partially answered	Report summarizes outputs but does not synthesize findings into coherent knowledge	Report absent, superficial, or does not address the central discovery question
Presentation	10-minute presentation is well-structured, findings are communicated clearly, and the team demonstrates full understanding of their results	Presentation covers key findings but structure or clarity could be improved, team handles questions adequately	Presentation is incomplete or difficult to follow, team struggles with some questions	Presentation is very poor, findings are not communicated, or team cannot answer basic questions

6. Cross-Group Activity: Mining Expo

At the end of the semester, all groups present at a Mining Expo. Because every group applied the same 5-phase methodology to a different banking dataset, the expo creates structured ground for comparison without unfair overlap.

Each group addresses the following questions during their presentation:

- Which association rules were the most surprising, and why?
- Which clustering method produced the most interpretable segments for your dataset?
- What anomalies were found, and what do they suggest in a real banking context?
- How do your findings compare to what another group discovered in a different banking domain?

The expo reinforces that data mining techniques are domain-agnostic, but the meaning of discovered knowledge is always shaped by context.